## PROYECTO ESPECIAL CHIRA-PIURA

DESCARGAS MEDIAS DIARIAS EN $M^3/S$. Y VOLUMENES DIARIOS

DEL RESERVORIO DE POECHOS.

**MES: ABRIL 2020**

| DIA | CANAL EJIDOS [$M^3/S$] | CANAL CHECA [$M^3/S$] | TUNEL [$M^3/S$] | ALIVIA-DERO [$M^3/S$] | TUNEL+ ALIVIAD. [$M^3/S$] | CANAL DERIV. [$M^3/S$] | CANAL HUAYPIRA [$M^3/S$] | EVAPO-RACION [$M^3/S$] | SALIDA TOTAL [$M^3/S$] | INGRESO [$M^3/S$] | BALANCE ($m^3/s$) (+) Q | BALANCE ($m^3/s$) (-) Q | VOL [MMC] | COTA [MSNM] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 34.83 | 15.6 | 20.58 | 0.0 | 20.6 | 55.0 | 0.25 | 3.53 | 79.4 | **152.3** | 72.9 | | 333.5 | 102.33 |
| 2 | 34.81 | 16.0 | 21.00 | 0.0 | 21.0 | 55.0 | 0.25 | 3.56 | 79.8 | **197.9** | 118.1 | | 339.8 | 102.47 |
| 3 | 34.81 | 16.0 | 19.92 | 0.0 | 19.9 | 57.9 | 0.25 | 3.59 | 81.7 | **165.0** | 83.3 | | 350.0 | 102.69 |
| 4 | 34.79 | 11.5 | 16.50 | 0.0 | 16.5 | 59.0 | 0.25 | 3.61 | 79.4 | **117.6** | 38.2 | | 357.2 | 102.84 |
| 5 | 34.22 | 10.3 | 15.32 | 0.0 | 15.3 | 55.7 | 0.40 | 3.63 | 75.1 | **126.0** | 50.9 | | 360.5 | 102.91 |
| 6 | 32.60 | 16.0 | 23.13 | 0.0 | 23.1 | 55.0 | 0.40 | 3.63 | 82.2 | **82.2** | | | 364.9 | 103.00 |
| 7 | 32.57 | 16.0 | 53.79 | 0.0 | 53.8 | 55.0 | 0.40 | 3.63 | 112.8 | **112.8** | | | 364.9 | 103.00 |
| 8 | 32.49 | 16.8 | 64.37 | 0.0 | 64.4 | 55.0 | 0.40 | 3.63 | 123.4 | **123.4** | | | 364.9 | 103.00 |
| 9 | 32.44 | 16.1 | 21.11 | 0.0 | 21.1 | 55.0 | 0.40 | 3.63 | 80.1 | **97.5** | 17.4 | | 364.9 | 103.00 |
| 10 | 32.43 | 16.0 | 21.00 | 0.0 | 21.0 | 55.0 | 0.40 | 3.68 | 80.1 | **91.7** | 11.6 | | 366.4 | 103.03 |
| 11 | 32.41 | 14.8 | 19.79 | 0.0 | 19.8 | 55.0 | 0.40 | 3.65 | 78.8 | **96.2** | 17.4 | | 367.4 | 103.05 |
| 12 | 32.57 | 14.0 | 19.00 | 0.0 | 19.0 | 55.0 | 0.40 | 3.65 | 78.1 | **94.3** | 16.2 | | 368.9 | 103.08 |
| 13 | 33.00 | 14.0 | 18.99 | 0.0 | 19.0 | 55.0 | 0.40 | 3.67 | 78.1 | **164.9** | 86.8 | | 370.3 | 103.11 |
| 14 | 32.99 | 14.0 | 44.50 | 60.4 | 104.9 | 57.1 | 0.40 | 3.70 | 166.1 | **308.5** | 142.4 | | 377.8 | 103.26 |
| 15 | 32.97 | 14.0 | 45.00 | 264.6 | 309.6 | 59.0 | 0.40 | 3.73 | 372.7 | **372.7** | | | 390.1 | 103.50 |
| 16 | 32.93 | 14.0 | 45.00 | 187.5 | 232.5 | 56.8 | 0.40 | 3.73 | 293.4 | **293.4** | | | 390.1 | 103.50 |
| 17 | 32.92 | 14.0 | 45.00 | 2.1 | 47.1 | 52.3 | 0.40 | 3.74 | 103.6 | **183.4** | 79.9 | | 390.1 | 103.50 |
| 18 | 32.92 | 14.0 | 51.67 | 0.0 | 51.7 | 46.0 | 0.40 | 3.76 | 101.8 | **144.7** | 42.8 | | 397.0 | 103.63 |
| 19 | 32.73 | 14.0 | 65.00 | 0.0 | 65.0 | 46.0 | 0.40 | 3.76 | 115.2 | **115.2** | | | 400.7 | 103.70 |
| 20 | 32.28 | 14.0 | 44.75 | 0.0 | 44.8 | 46.0 | 0.40 | 3.77 | 94.9 | **113.4** | 18.5 | | 400.7 | 103.70 |
| 21 | 32.29 | 14.2 | 17.09 | 0.0 | 17.1 | 44.3 | 0.40 | 3.77 | 65.6 | **109.6** | 44.0 | | 402.3 | 103.73 |
| 22 | 32.16 | 14.5 | 19.50 | 0.0 | 19.5 | 40.0 | 0.40 | 3.79 | 63.7 | **82.2** | 18.5 | | 406.1 | 103.80 |
| 23 | 32.10 | 14.5 | 19.50 | 0.0 | 19.5 | 40.0 | 0.40 | 3.79 | 63.7 | **69.5** | 5.8 | | 407.7 | 103.83 |
| 24 | 32.10 | 14.5 | 20.75 | 0.0 | 20.8 | 40.0 | 0.40 | 3.79 | 64.9 | **90.4** | 25.5 | | 408.2 | 103.84 |
| 25 | 32.08 | 14.5 | 22.49 | 0.0 | 22.5 | 40.0 | 0.40 | 3.80 | 66.7 | **85.2** | 18.5 | | 410.4 | 103.88 |
| 26 | 31.86 | 14.3 | 36.33 | 0.0 | 36.3 | 44.4 | 0.40 | 3.81 | 85.0 | **142.8** | 57.9 | | 412.0 | 103.91 |
| 27 | 31.51 | 14.0 | 62.71 | 0.0 | 62.7 | 48.0 | 0.40 | 3.82 | 114.9 | **114.9** | | | 417.0 | 104.00 |
| 28 | 32.23 | 14.0 | 38.75 | 0.0 | 38.8 | 48.0 | 0.40 | 3.82 | 91.0 | **91.0** | | | 417.0 | 104.00 |
| 29 | 32.19 | 14.0 | 23.73 | 0.0 | 23.7 | 48.0 | 0.40 | 3.82 | 76.0 | **81.7** | 5.8 | | 417.0 | 104.00 |
| 30 | 32.19 | 15.0 | 35.15 | 0.0 | 35.2 | 48.0 | 0.40 | 3.82 | 87.4 | **81.6** | | -5.8 | 417.5 | 104.01 |
| 01/05 | | | | | | | | | | | | | 417.0 | 104.00 |
| SUMA | 984.42 | 434.70 | 971.42 | 514.58 | 1,486.00 | 1,526.58 | 11.40 | 111.31 | 3135.3 | 4101.7 | 972.2 | -5.8 | | |
| PROM | 32.81 | 14.49 | 32.38 | 17.15 | 49.53 | 50.89 | 0.38 | 3.71 | 104.5 | 136.7 | 46.3 | -5.8 | 385.6 | 103.40 |
| MAX | 34.83 | 16.83 | 65.00 | 264.58 | 309.58 | 59.00 | 0.40 | 3.82 | 372.7 | 372.7 | 142.4 | -5.8 | 417.5 | 104.01 |
| MIN | 31.51 | 10.33 | 15.32 | 0.00 | 15.32 | 40.00 | 0.25 | 3.53 | 63.7 | 69.5 | 5.8 | -5.8 | 333.5 | 102.33 |
| VOLUMEN | 85.05 | 37.56 | 83.93 | 44.46 | 128.39 | 131.90 | 0.98 | 9.62 | 270.9 | 354.4 | 84.0 | -0.5 | | |
| Proyectado al 30 de Abril (MMC) | | | | | | | | | 270.9 | 354.4 | | 83.5 | **417.0** MMC | |
| Volumen Programado | | | | | | | | | | **303.7** MMC Q= | | 117.2 | $m^3/s$. | |

116.68 %

## PROYECTO ESPECIAL CHIRA PIURA

## RIO CHIRA: VOLUMENES MENSUALES EN MMC.

| | Ene | Feb | Mar | Abr | May | Jun | Jul | Ago | Sep | Oct | Nov | Dic | Anual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| OFERTA RIO CHIRA (MMC) | 126.22 | 484.22 | 396.81 | 303.72 | 385.13 | 355.93 | 231.09 | | | | | | **2,283.1** |
| Q($m^3/s$) | 47.1 | 193.3 | 148.2 | 117.2 | 143.8 | 137.3 | 86.3 | | | | | | |
| Aporte Real / Oferta (%) | 171.0 | 63.0 | 73.5 | 116.7 | | | | | | | | | **424.1** |
| **Aporte Real (MMC)** | 215.8 | 304.9 | 291.6 | 354.4 | | | | | | | | | **1,166.8** |
| **Aporte Acum. (MMC)** | 215.8 | 520.8 | 812.4 | 1,166.8 | | | | | | | | | |
| **SALIDA TOTAL. (MMC)** | 172.0 | 310.7 | 241.1 | 270.9 | | | | | | | | | **994.8** |

# PROYECTO ESPECIAL CHIRA-PIURA

DESCARGAS MEDIAS DIARIAS

DE LOS RIOS CHIRA Y PIURA ($M^3/S$)

**MES: ABRIL 2020**

| DIA | PUENTE INTERN. [$M^3/S$] | EL CIRUELO [$M^3/S$] | PARAJE GRANDE [$M^3/S$] | PRESA SULLANA | | | PUENTE R. PIURA (1) [$M^3/S$] | CURUMUY [$M^3/S$] | PRESA EJIDOS | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | R.Chira [$M^3/S$] | C. Norte [$M^3/S$] | TOTAL [$M^3/S$] | | | C. EJIDOS [$M^3/S$] | R. PIURA (2) [$M^3/S$] |
| 1 | 44.5 | 156.3 | 9.6 | 5.2 | 17.8 | 23.0 | 0.0 | 30.8 | 34.83 | 4.0 |
| 2 | 126.4 | 281.3 | 21.2 | 7.4 | 17.8 | 25.2 | 0.0 | 31.6 | 34.81 | 3.2 |
| 3 | 67.0 | 215.1 | 15.1 | 11.8 | 17.8 | 29.6 | 0.0 | 36.1 | 34.81 | -1.3 |
| 4 | 57.8 | 166.8 | 8.9 | 9.0 | 17.8 | 26.8 | 1.9 | 35.5 | 34.79 | 1.2 |
| 5 | 51.3 | 155.9 | 9.1 | 8.0 | 17.8 | 25.8 | 12.7 | 35.0 | 34.22 | 12.0 |
| 6 | 43.0 | 149.4 | 9.1 | 8.1 | 16.7 | 24.8 | 7.6 | 32.5 | 32.60 | 7.6 |
| 7 | 49.8 | 155.6 | 8.0 | 26.3 | 16.7 | 43.0 | 2.9 | 32.4 | 32.57 | 3.1 |
| 8 | 54.6 | 155.0 | 8.7 | 59.8 | 17.1 | 76.8 | 2.1 | 32.4 | 32.49 | 2.2 |
| 9 | 47.5 | 140.7 | 7.5 | 10.0 | 17.3 | 27.3 | 2.0 | 33.3 | 32.44 | 1.2 |
| 10 | 48.1 | 153.6 | 7.3 | 6.8 | 17.3 | 24.1 | 1.5 | 33.1 | 32.43 | 0.8 |
| 11 | 35.0 | 113.2 | 7.5 | 6.8 | 17.3 | 24.1 | 0.0 | 32.4 | 32.41 | 0.0 |
| 12 | 34.0 | 118.3 | 8.2 | 6.0 | 16.7 | 22.8 | 3.1 | 34.3 | 32.6 | 1.3 |
| 13 | 107.4 | 226.9 | 12.6 | 6.0 | 16.1 | 22.1 | 1.5 | 31.8 | 33.0 | 2.7 |
| 14 | 169.7 | 420.4 | 17.7 | 32.9 | 16.0 | 48.9 | 0.0 | 31.4 | 33.0 | 1.6 |
| 15 | 162.9 | 377.1 | 17.6 | 250.2 | 16.1 | 266.3 | 4.6 | 35.0 | 33.0 | 2.6 |
| 16 | 100.9 | 241.4 | 8.8 | 266.5 | 14.5 | 281.0 | 5.2 | 32.8 | 32.9 | 5.4 |
| 17 | 102.7 | 256.7 | 15.6 | 48.5 | 15.8 | 64.3 | 51.0 | 30.2 | 32.9 | 53.7 |
| 18 | 81.7 | 191.4 | 18.3 | 37.3 | 15.5 | 52.8 | 60.3 | 24.4 | 32.9 | 68.9 |
| 19 | 70.0 | 168.2 | 17.3 | 50.6 | 15.6 | 66.2 | 42.0 | 25.0 | 32.7 | 49.7 |
| 20 | 60.3 | 144.4 | 16.4 | 42.0 | 16.6 | 58.6 | 29.6 | 25.2 | 32.3 | 36.6 |
| 21 | 47.1 | 126.0 | 16.6 | 6.8 | 16.6 | 23.4 | 25.3 | 24.4 | 32.3 | 33.2 |
| 22 | 49.4 | 113.3 | 9.9 | 5.7 | 16.6 | 22.3 | 13.5 | 20.1 | 32.2 | 25.6 |
| 23 | 47.6 | 104.6 | 9.4 | 5.6 | 16.6 | 22.2 | 5.1 | 18.7 | 32.1 | 18.5 |
| 24 | 47.5 | 99.2 | 8.7 | 6.3 | 16.6 | 22.9 | 0.4 | 18.6 | 32.1 | 13.9 |
| 25 | 56.7 | 114.3 | 8.6 | 6.0 | 15.6 | 21.6 | 0.0 | 17.9 | 32.1 | 14.2 |
| 26 | 100.1 | 192.0 | 8.3 | 9.3 | 15.6 | 24.9 | 0.0 | 18.8 | 31.9 | 13.1 |
| 27 | 54.3 | 122.5 | 8.2 | 47.1 | 15.7 | 62.8 | 4.2 | 23.9 | 31.6 | 11.9 |
| 28 | 51.1 | 115.8 | 8.2 | 27.0 | 16.1 | 43.0 | 0.0 | 25.4 | 32.2 | 6.9 |
| 29 | 49.0 | 108.4 | 7.9 | 10.3 | 16.1 | 26.4 | 0.0 | 25.1 | 32.2 | 7.1 |
| 30 | 42.8 | 93.1 | 7.7 | 21.0 | 16.1 | 37.1 | 0.0 | 25.4 | 32.2 | 6.8 |
| SUMA | 2060.1 | 5176.7 | 338.1 | 1044.1 | 495.95 | 1540.00 | 276.39 | 853.42 | 984.5 | 407.45 |
| PROM | 68.7 | 172.6 | 11.3 | 34.8 | 16.53 | 51.33 | 9.21 | 28.45 | 32.8 | 13.58 |
| MAX | 169.7 | 420.4 | 21.2 | 266.5 | 17.80 | 280.98 | 60.29 | 36.13 | 34.8 | 68.85 |
| MIN | 34.0 | 93.1 | 7.3 | 5.2 | 14.48 | 21.60 | 0.00 | 17.93 | 31.6 | -1.33 |
| VOLUMEN | 178.0 | 447.3 | 29.2 | 90.2 | 42.85 | 133.06 | 23.88 | 73.74 | 85.1 | 35.20 |
| NORMAL | 90.0 | 215.7 | 32.1 | 90.8 | | 133.1 | | | | |
| ANOMALIA | -21.3 | -43.1 | -20.8 | Poechos | | Sullana | | | | |

( 1 ) PUENTE SANCHEZ CERRO

( 2 ) ESTACION EL PAPAYO

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| % | 76.30 | 80.00 | 35.11 | | | | **Avda:** | **27.6** | 266.9 | 276.5 | **37.2** |

## Proyecto Chira-Piura

PRECIPITACIONES (mm)

**MES: ABRIL 2020**

| DIA | PTE INT. mm | EL CIR. mm | PJE GDE mm | Ardilla mm | Alivia dero mm | Sullana mm | Check 29+900 mm | CPTO PIURA mm | PRESA EJIDOS mm | Chulu canas mm | Tbo Gde mm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 15.0 | 6.8 | 4.0 | 0.3 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.3 | 0.8 |
| 2 | 38.8 | 1.7 | 18.5 | 15.7 | 11.0 | 5.2 | 0.8 | 1.4 | 4.8 | 8.8 | 19.0 |
| 3 | 0.0 | 0.0 | 3.0 | 0.3 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 4 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 6 | 3.7 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 7 | 1.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.4 |
| 8 | 3.8 | 15.2 | 8.5 | 0.6 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.7 |
| 9 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 10 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 11 | 2.7 | 0.7 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 12 | 0.0 | 0.6 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 13 | 9.0 | 7.8 | 13.5 | 1.6 | 0.9 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 14 | 6.5 | 0.6 | 39.0 | 6.8 | 2.5 | 0.0 | 0.0 | 0.0 | 0.0 | 10.0 | 4.5 |
| 15 | 0.0 | 0.8 | 12.5 | 6.7 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.0 |
| 16 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 17 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 18 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 19 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 20 | 0.0 | 0.0 | 8.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 21 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 22 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 23 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 24 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 25 | 9.2 | 5.8 | 2.0 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 26 | 0.5 | 0.0 | 6.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 27 | 0.0 | 0.0 | 3.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 28 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 29 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 30 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| **Total Mensual** | 90.3 | 40.0 | 119.0 | 33.0 | 14.4 | 6.7 | 0.8 | 1.4 | 4.8 | 20.2 | 26.4 |
| PROM | 3.0 | 1.3 | 4.0 | 1.1 | 0.5 | 0.2 | 0.0 | 0.0 | 0.2 | 0.7 | 0.9 |
| MAX | 38.8 | 15.2 | 39.0 | 15.7 | 11.0 | 5.2 | 0.8 | 1.4 | 4.8 | 10.0 | 19.0 |
| MIN | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| NORMAL | 110.9 | 123.0 | 118.2 | 113.2 | 18.2 | 17.6 | 7.2 | 9.9 | | 27.2 | 13.6 |
| ANOMALIA | -20.6 | -83.0 | 0.8 | -80.2 | -3.8 | -10.9 | -6.4 | -8.5 | | -7.0 | 12.8 |
| % | 81.42 | 32.52 | 100.68 | 29.15 | 79.12 | 38.07 | 11.11 | 14.14 | | 74.26 | 194.12 |
| Promedio | 65.46 | | | | | | | | | | |